

Challenges for copyright in a digital age

I D Bramhill and M R C Sims

The act of breaching copyright is probably the most common civil offence, and is often not considered as being unlawful by the perpetrator. The revenue that is lost by a copyright owner when illegal copying takes place can be significant. Losses may become unacceptably large in the future given the expected explosion of computerised, multimedia services. This paper discusses the problems copyright owners face when trying to maintain cost-effective control of their copyright in a digital age. It then proposes an initial model of a software-based system that provides copyright protection of multimedia information when delivered by Internet-based services.

1. Introduction

Authors and artists have certain rights when they produce a work-of-art — these rights are automatically assigned to them and no registration is needed. These include copy rights. When a copy of a work-of-art is made some fee can lawfully be claimed by the author for its use. If an author¹ finds evidence that someone is making copies of his work-of-art without permission he can take the infringer to a court of law and reclaim lost revenue. Authors would naturally like to maximise the amount of revenue that comes to them from others making copies of their work-of-art with a minimum amount of effort on their own part. Copyright infringement is a civil offence and so the onus is on the author to protect his work-of-art. Traditionally this has been easy due to the physical nature of works-of-art — generally it costs less and is more desirable to buy a high-quality copy of a book from a store than to make an illegal copy. Due to the increase in electronic distribution of information and the reduction in cost of storage of such information, copyright infringement is increasing. New mechanisms are needed to ensure that authors preserve their revenue stream.

1.1 Copyright and copy protection

Data that is in the digital domain can be reproduced, error free, with as little effort as a 'drag-and-drop' operation using a graphical user interface. If the same process is repeated on the first generation copy the result is a perfect second-generation copy. An equivalent process in the analogue domain would be to repeatedly use a photocopier on its own output, but this results in a rapid reduction in quality after a small number of copy generations. When we

work in the digital domain we have the ability to pass on a perfect copy to anyone, anywhere in the world.

The great benefit for a recipient of digital data is the increase in quality of the copy that he receives. Some of the benefits for the sender are that he can provide a better service, to more people, in less time, and at a fraction of the cost. Because the material is in digital form there is less physical protection available for it and so copyright owners² have lost some of the control they once had. This reduces the amount of revenue that they can collect. The international laws for copyright give the owner the right to make a charge for the supply of a copy of an original work-of-art. Therefore owners want to encourage copies to be made of their work-of-art to increase their revenue. Copyright owners also want to be able to control the copies once they have been made in order to protect future revenue. A copy protection system provides them with a method of **controlling** copies, it does not attempt to **prevent** copies from being made, because this is not possible.

Being unable to prevent a copy from being made would not be a problem if one could detect the act of copying. If this were the case, then copyright owners could still collect revenue when their works-of-art are used.

Unfortunately it is not feasible to detect the act of making a copy when it matters, that is, when combating organised piracy. For example, the digital information to be

¹ The author/owner/user, depicted as male throughout this paper, could equally be female.

² Copyright owner: a person, or organisation, that owns the copyright for a work-of-art (which can be a piece of text, music, painting, film). The copyright owner can be someone other than the author of a work-of-art, e.g. Michael Jackson is the owner of the copyright of many works-of-art of which Lennon and McCartney are the authors. Sometimes this paper shortens the term 'copyright owner' to just 'owner'.

copied can always be put into a computer that is not connected to any network, and that has no kind of communications capability. Although the initial recovery of the information can be detected and charged for, once put into the pirate computer the process of making multiple copies cannot be detected.

1.2 The impact of new technology

Copyright-protected material is starting to be provided by many new delivery methods; this makes it susceptible to new threats. An example of such a new delivery method is Digital Video Disc (DVD) which had its world market launch delayed by concern over copyright issues:

- ‘..the studios have said... that no titles will be released until all the outstanding copy protection issues have been resolved to the satisfaction of all parties’ [1],
- ‘..everyone agrees that copy protection is the most visible issue. The movie industry has steadfastly upheld their intention to withhold publishing titles until they are convinced there is an acceptable means for protecting their assets from being copied. The method of copy protection used, they insist, must also be applied to computers. Therein lies the problem.’ [2].

The DVD format allows 133 minutes of broadcast quality video and sound to be held on a disc that has the same physical dimensions as a music CD (compact disc). DVD is sometimes called Digital *Versatile* Disc because it can carry any information, not just video. It is therefore expected to be of significant interest to computer manufacturers who see it as providing a step change in the capacities available with CD-Read Only Memory (CD-ROM) giving DVD-ROM.

Initial capacity for a DVD-ROM will be 4.7 gigabytes, rising to a capacity of 17 gigabytes for double sided, dual-layer technology. It is expected that DVD-recordable drives will soon appear. A date of mid-1997 is currently predicted [3] and machine prices, when driven by a powerful computer industry, will quickly fall to be similar to that of CD-recordable drives (currently found for less than US\$2000). It can therefore be seen that perfect copies of works in which the film industry have literally invested billions of American dollars, will now be available as source material for ‘professional’ and ‘home’ pirating using personal computers.

1.3 DVD protection

The agreed industry-wide mechanism for the protection of copyright in the DVD system comprises a number of techniques as described below.

Each DVD player and disc pressing will be supplied with one of six regional codes, this will ensure that a disc that is released in North America will not function on a DVD player that is bought in Europe.

Some of the digital information will be protected using a process called encryption³. The specific implementation will be licensed by a governing body so that the manufacture of DVD players can be controlled. The movie soundtrack and imagery will be encrypted as two separate streams of information (see Fig 1). The DVD player uses its licensed technology to access information on the disc that tells it how to decrypt the streams.

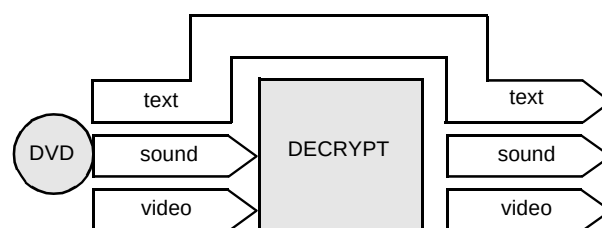


Fig 1 DVD protection.

A DVD player will not have connectors that give access to the decrypted digital information. The analogue video signal that is output will be protected by a technique owned by a company called Macrovision. This technique causes recorded video quality to be reduced.

It is hoped that laws will be imposed to make it illegal to sell or possess technology that tries to circumvent the copyright protection mechanism for the DVD system. An alternative approach is for industry to also design the circumvention technology and to patent it. This approach has the benefit that someone building and selling circumvention technology can be sued for patent violation and this is more easily achieved internationally in comparison to proving copyright violation.

1.4 Identifying requirements

In the DVD example requirements for the copyright protection mechanism came from the owners identifying the environment in which the work is going to be used. The environment is that of traditional retail distribution, selling physical items to anonymous customers. This paper considers a different environment having its own requirements and which possibly poses the greatest challenge to

³ Encryption is a process that is part of an area called cryptography. We make reference to these areas many times in this paper and so will give a definition of some of the terms; greater depth of the subject can be found in Phoenix [4]. ‘A cipher is a secret method of writing, whereby plaintext... is transformed into ciphertext. The process of transforming plaintext into ciphertext is called ... encryption; the reverse process ... is called ... decryption. Both encryption and decryption are controlled by a cryptographic key.’ [5].

copyright in the digital age. The environment is that of providing works-of-art using Internet-based services.

There are a number of schemes published or currently in use such as that proposed by Choudhury et al [6] and Adobe Acrobat® [7]. This paper proposes alternatives to some of the published mechanisms and extends others. It also combines mechanisms to create a complete end-to-end solution.

This environment provides a significant challenge because owners wish to supply digital information on a world-wide basis into many individual environments where they have no direct control, a situation unlike that of DVD. The copyrighted material is therefore supplied to a user⁴ by a copy protection system, but once on a user's personal computer he can make many attempts to subvert the copy protection system and so resell the material as his own over the Internet. Should the user find a significant weakness in the protection mechanism he could even render the entire system useless by publicising the weakness.

In addition to the requirements imposed by using the model of Internet-based services there are a number of requirements that must be satisfied if a copy protection mechanism is to be attractive to users and owners. For example, the owners do not want to have to perform many processes to gain an adequate level of protection. Also, users do not want to have to go through a registration phase each time they want access to information from a new source. Such requirements were considered when developing our proposed mechanism.

2. Can prevention work?

2.1 An example of extreme prevention

Consider a work-of-art that takes the form of written text that is the original manuscript of a poem. The owner can easily prevent copyright infringement by never allowing any user to read the poem, and this can be ensured by keeping it locked in a security vault. The obvious problem here is that this approach will not result in the collection of any revenue, unless he charges a fee for access to the vault.

For our fictional owner it appears that he is still maintaining control, but there is a problem. The user can easily reproduce the poem from memory and resell it once he leaves the vault. If the work-of-art is a novel the user can read it aloud into an audio recorder and reproduce it later, or even memorise it. The owner could make a restriction that no manner of copy technology, such as an audio recorder or camera, ever leaves the vault but he is still faced with the problem of a user with a photographic memory.

This copy protection system has a very high level of control but we can see from it that no copy protection

system can justly make the claim of being absolutely secure. With the additional requirement of checking for users possessing recording devices, the amount of revenue that could be lost has been reduced to users having photographic memories. Each time a decision is made on how a copy protection system functions we must consider how to minimise the possible threats, and must decide if the maximum potential fraud is below the threshold of what is considered an unacceptable loss to the owner.

2.2 Cryptography to the rescue

Now we consider the scenario of the owner needing to take the manuscript out of the vault to allow access by a user who cannot physically get to it. The owner can take the original manuscript to the user but if the manuscript is stolen in transit then the owner will have lost the work-of-art. The owner can make a copy of the manuscript and take that to the user; but if the copy is stolen, he still suffers a significant loss. The owner is faced with a similar problem to that of a government wanting to send a message containing secret information to their spy in another country. If the message is intercepted by an enemy, then he should not be able to determine the secret information. Governments have long achieved such protection by use of cryptography 'which embraces methods for rendering data unintelligible to unauthorised parties' [8].

By using an encryption process the poem owner can create a copy of his poem as a ciphertext version which he takes to the user (see Fig 2). The owner uses the matching decryption process to recover a plaintext copy of the original which they show to the user. Here cryptography is being used to provide confidentiality; if the owner loses the ciphertext, no one can recover the original plaintext of the poem without access to the decryption process and the cryptographic key.

The fictitious owner has protected his poem in transit, but is faced with another problem. As soon as the

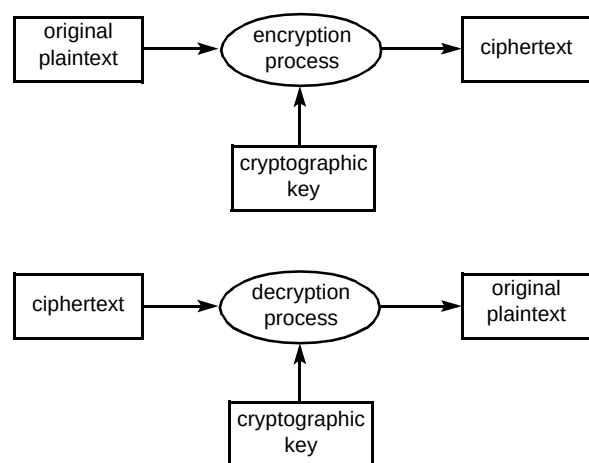


Fig 2 Cryptography terminology.

⁴ A person, or organisation who makes use of a copyrighted work-of-art.

decryption process⁵ is applied, the plaintext is revealed, and so the amount of control he has over the work is again reduced. The poem's owner can maintain control by never leaving the plaintext unattended, but this is not possible when network delivery is involved.

2.3 Trust in cryptography

Cryptography is normally used to protect the transfer of information between two parties that trust each other.

For example, our fictitious government encrypts plaintext and sends the ciphertext to their spy. The spy knows the decryption process and the cryptographic key and recovers the plaintext message. Once the spy has read the plaintext message, it is destroyed to ensure the spy's own safety.

Commercial companies use cryptography to protect information sent by computers operating between departments. These departments can trust that each will not disclose the decryption process or the cryptographic key.

When using cryptography in an Internet-based copy-protection mechanism, we cannot consider the computer of the user to be trustworthy. We must regard all of the users as potential pirates because we know there are a small number of real pirates trying to defraud the system. Figure 3 shows how the areas that the sender trusts alter between the traditional use of cryptography (a) and a situation where the recipient is untrustworthy (b).

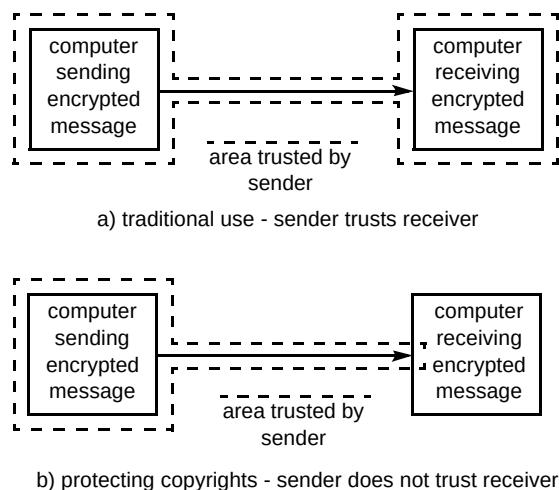


Fig 3 Trust in cryptography.

⁵ For the decryption process to succeed the cryptographic key must be the correct key. We must ensure that all parties who should have the correct key have it when they need it, and must ensure that any party who should not have the key cannot obtain it without expending a significant level of effort. To do this requires 'cryptographic key management' which is a large subject area and is not covered by this paper.

2.4 Making a viable solution using cryptography

If we are to use cryptography to protect information in an environment of limited trust, we must limit access to the plaintext. Figure 4 shows the journey a work-of-art takes to get to the brain of the user. At some point in this journey the decryption must happen, i.e. the work-of-art must leave the area trusted by its owner. This should happen as late as possible to make it difficult for a fraudster to get control of the plaintext.

Ideally we would like to perform the decryption when the encrypted work-of-art is in the brain of the user but this idea is clearly unachievable and unacceptable. The next best approach is to decrypt the information just before it reaches the eyes and ears of the user.

This can be achieved by requiring the user to have a special device that performs the decryption process but which is configurable only by the sender. This device must also control the use of the work-of-art, so that without it the work-of-art is unusable⁶. For example, to prevent quality copies of a video tape being made, a distributor could supply the user with a tape that is encrypted such that it can only be played in that user's video player. The video player must be tamper-proof, must incorporate a television screen and provide no means of attachment to any other recording device.

Clearly this would be an expensive way to distribute all video tapes, but could possibly be a solution when a small number of users require information of high value. Because the decryption process now occurs in hardware and not in the brain, fraudsters⁷ have an opportunity to make copies, for example, by using a video camera to record the images on the screen. The quality of the copies made using such a technique would probably be so low as not to pose a significant threat to the copyright owner.

Any solution that requires expensive devices at the user's machine will be limited to special applications. This paper considers a model of special applications provided by Internet-based services to a world-wide client base. To reduce the costs of a copy-protection system to a level that is acceptable to both owners and users we consider that only a software-based copy protection mechanism utilising cryptography will be commercially viable. We therefore need to consider the problems that this may present.

⁶ In some cases an owner may want to control the use of the work-of-art. For instance, an owner of a journal may wish to control how it is viewed, if it can be printed to paper, and the level of quality when printed.

⁷ A fraudster is a person, or organisation, who attempts to make unauthorised copies for financial gain or attempts to provide users with the means to make unauthorised copies. A fraudster will usually be a user.

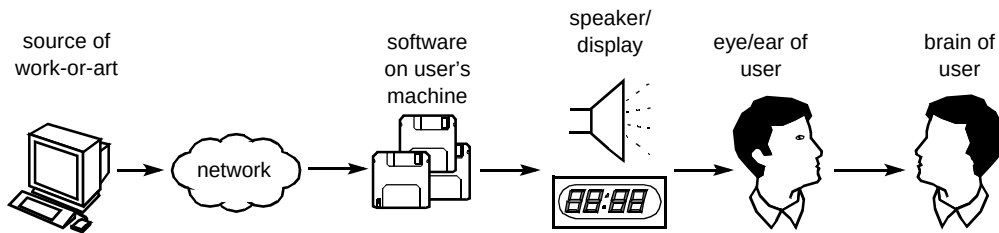


Fig 4 Where to decrypt.

3. Challenges for a software-only solution

3.1 Binding software to a user

The first problem to be considered is that a software-only solution is subject to all of the problems faced by information in the digital domain, i.e. its elements can be easily copied and distributed. We therefore need to ensure that a duplicate of the copy protection software of one user is of no use to anyone other than that user.

This implies the need for a strong⁸ one-to-one binding between the software of a user, and the user. The binding could be achieved by making it undesirable for users to distribute copies of the software, e.g. by making it display sensitive information about the user, such as their home address or other personal information. However, this would probably be unacceptable to users.

The best way to achieve the binding is to use smart cards⁹. Smart cards have been designed to provide strong identification of a person, so the strong binding we require can be achieved by binding the software to the smart card, and the smart card to the user. Although smart cards provide a secure cryptographic environment they should not be used to decrypt the work-of-art. If smart cards were used in this way, a fraudster would only need to intercept the plaintext output to access the work-of-art (see Fig 5). Unfortunately the smart cards also need to be distributed to the users before the work-of-art can be used; this may be unacceptable in some circumstances.

3.2 Binding software to a machine

Another way to achieve the strong binding we require is to bind the decryption software to the computer on which it is run or the terminal from which it is used. Biometrics identify a human to a high level of probability. A number of characteristics are measured to do this. Similarly, a number of characteristics of a computer can be measured to achieve a similar level of probability of its identity, which we call a 'cybermetric'. Examples of such characteristics are:

- the physical components which the computer comprises (size of memory, presence of CD drive),
- characteristics of the physical components (manufacturer, number of tracks on a hard disk),
- location of static information on a hard disk (bad sectors),
- location of long-lived files on a hard disk (operating system executables),
- operational characteristics,
- logical directory and file structures,
- files specifically created to identify the machine,
- data added to long-lived files to identify the machine,
- the configuration of applications and the operating system.

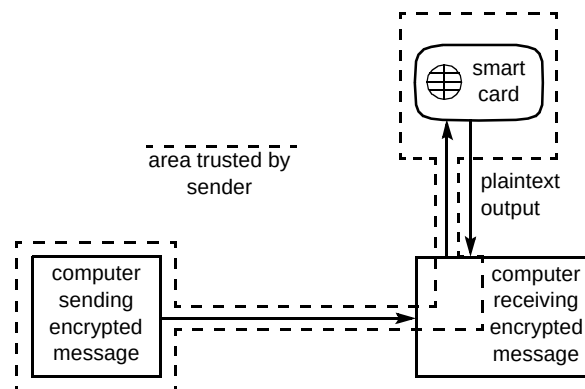


Fig 5 Smart card decryption.

For fraudsters to make use of a software decryption process belonging to another user, they would have to recreate both the logical and physical characteristics of that machine, and the characteristics of the way the original user operated all of the separate applications.

3.3 Binding a work-of-art to a user

In the earlier example of the tamper-proof video player, there was a one-to-one relationship between an instance of a work-of-art and the special viewing equipment. If a user gave their video tape to another user with a similar device then his video tape could not be played on the second user's equipment.

⁸ Strong means that it cannot be easily tampered with, broken or forged.

⁹ Smart cards are usually credit card sized devices made of plastic that have micro-electronic circuits embedded within. The circuitry provides a secure, tamper-resistant computing environment that can be used to implement cryptographic mechanisms.

This approach also needs to be taken with a software-only copy-protection mechanism. When information is given to the user, the owner needs to control that information. If the work-of-art is stored in a manner that can be read only by the software of the user to whom it was sold then control has been maintained. This can be achieved by using a different cryptographic key in every software instance.

3.4 Reverse engineering

The major weakness of all software processes is that a fraudster can take apart the software using reverse engineering techniques; we recognise that this cannot be prevented. The intention is to engineer the copy-protection mechanism in such a way as to make the act of breaking it uneconomical for a fraudster, i.e. the cost is larger than the cost of compliance.

Once reverse-engineering techniques are used to find how one instance of the software functions, the entire software system is broken since all instances work the same way. Ideally what is required is that such an attack only breaks that one instance of the software, limiting the fraud potential to that user. What is required is the ability to produce software that provides identical functionality in every instance, but which provides this functionality in a unique manner for each instance. This is not achievable with current technology but will be the subject of further research.

What may be achievable is to produce the same functionality in a limited number of instances and to repeatedly clone each of these functional instance to give a large total population. It may then be acceptable to randomly distribute the clones around the world if the number of clones of any particular functional instance is small compared to the overall population size. When a clone of a functional instance is broken, the impact is limited to a fraction of the total population; the fraudster will have to track down the other clones of the same functional instance without being discovered.

The worst case scenario is that all of the users of the functional instance are informed of the illegal method, and all users convert to the illegal method of access. The number of functional instances that are required needs to be calculated by reviewing the acceptable level of fraud and this figure may be small, thus making the approach feasible.

3.5 Trust

Each clone of each instance must ensure the protection of the information with which it deals. For example, the owner of a work-of-art may wish for each clone to control the action of printing. If this control is not implemented because of an unintentional error in the clone, or because of

deliberate intent, then an unacceptable level of fraud could result. When supplying copyrighted information to a software copy-protection mechanism the owner must be assured that the software can be trusted to correctly implement the specified scheme and do nothing else.

4. Detecting illegal copies

4.1 Expecting the worst

At this point we recall earlier statements on the impossibility of creating a perfect system, and that software can be broken given enough resources. Both result in loss of control of the work-of-art by the owner. When a fraudster breaks the system we can accept that control has been lost or we can look for techniques that allow a degree of control to be maintained.

When an owner is made aware of an illegal copy he will have to prove to a legal system that he is the original owner and that his copyright has been infringed. If the illegal copy is a pirate music CD of a well-known recording artist the proof is likely to be simple but the owner has no way of knowing who supplied the original CD. If the owner could discover the source of the forgery he could deny that user access to future work. This is clearly not possible with physical objects that are available throughout the world, but what of information that is supplied by an Internet-based service?

4.2 Steganography

As we cannot detect the act of making a copy we want to be able to identify which user let their legally obtained copy be illegally copied. This can be achieved by detection schemes [9] that rely heavily upon a field of expertise referred to as 'steganography'. Schneier [10] gives this definition: 'Steganography serves to hide secret messages in other messages, such that the secret's very existence is concealed', in other words the secret message is subliminal.

An example of steganography is a message between two spies, printed in the personal column of a newspaper. The message, 'Hamlet, meet me at midnight on the castle battlements, Your Late Father,' is understandable to all readers, but it only has a hidden meaning between the two who know the code. For example Hamlet might know that 'castle battlements' indicates that the meeting should really take place on the bridge over the river, and the word 'late' means that one hour should be added to the time of 'midnight'.

4.3 Digital watermarks

Steganography is the underlying technology of digital watermarking. Watermarking a piece of paper takes place at the time of manufacture, but a digital watermark can be

embedded at any time. It is usually required that a digital watermark should not intrude on the normal use of the work-of-art.

To show how a digital watermark might be accomplished consider how a digital recording is made. The analogue sound is sampled at fixed intervals, and the frequency is represented as a number. The number can be stored in binary form, transferred electronically and then a close approximation of the sound is recreated by digital-to-analogue circuitry. The system works because the human ear is not capable of recognising the quantisation errors in the process. This means that a change in the least significant bit (LSB) cannot be noticed by the human ear, and by altering some of the least significant bits of digital words in a pre-arranged manner we can uniquely 'watermark' each recording [11].

An electronic version of an image represents the real one by quantization just as a sound signal, but instead of sampling frequencies the amount of red, green and blue at a small point (a pixel) are sampled. Each of the three colours is often allocated 8 bits of information, to recreate the colour of a pixel requires all 24 bits to be recombined. If the least significant bit of an 8-bit value is changed, this gives a very small change in the rendered colour and this change is not detectable to human sight. Image information has been lost but it can now be used for a watermark. If a maximum limit (10% is common) is placed on the number of pixels that are changed the overall effect will not be detected by viewing, but provides a significant bandwidth for a watermark.

This approach works because, to a human eye, the level of signal represented by the low-order bit does not carry any representable information; therefore the bit is redundant. Figures 6 — 9 show a simple way to use this redundancy to store a watermark in an image. Figure 6 shows an image to be watermarked. Figure 7 shows the LSBs of the red components of each of the pixels (black corresponds to a one, and white to a zero). If these bits are replaced with the watermark shown in Fig 8, the result is the image in Fig 9. To recover the watermark, the LSBs of the red components of Fig 9 are examined ¹⁰.

4.4 The lack of redundancy

The similarity of Figs 6 and 9 show that the bits that were replaced were redundant in the original image. This is the key element in how steganographic methods often function; they require redundancy (or noise) to be present. For this reason it is very difficult to apply such techniques to information representations with little or no redundancy, such as character codes. If a change is made to the least significant bit of a character code, the change results in a

new character being represented, and this can clearly be detected. Although such a change could be explained as a badly spelt word, the bandwidth is obviously very low. An increase in the bandwidth can be achieved by changing the words using a thesaurus, modifying punctuation within the accepted limits of the language, and by recasting sentences; all of these allow the same meaning to be conveyed in a different way. These approaches are not possible when the work being used as the carrier has no redundancy, an example of this is a novel; any change would be unacceptable to the author and obvious to students of it.



Fig 6 Unmodified image.

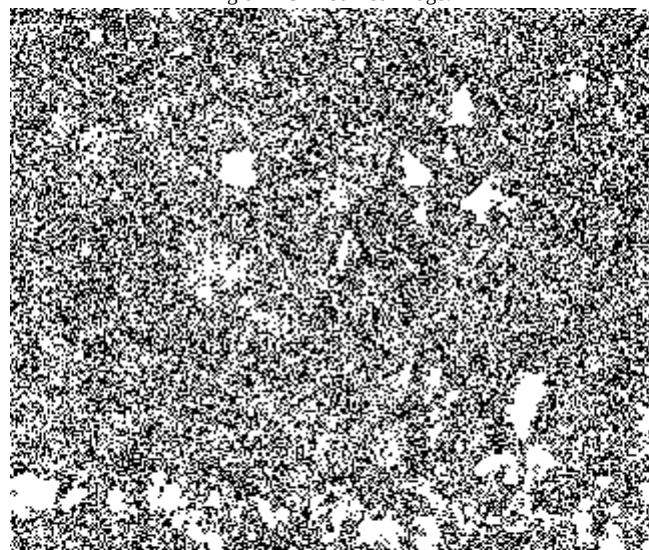


Fig 7 LSBs of unmodified image.

Bramhill and Sims. BT Laboratories

Fig 8 Embedded watermark.

¹⁰ This technique can easily be extended so that the original image is needed to recover the watermark.



Fig 9 Watermarked image.

Three ideas from AT&T [12—14] describe the techniques of ‘word shifting’, ‘line shifting’ and ‘character coding’ — these allow steganography to be used in difficult circumstances. With the first two techniques words or lines of text are shifted in relation to other words and lines by an amount smaller than that detectable by a human. To detect if a line has been shifted the preceding and following lines must be kept static, to detect if a word has been shifted the preceding and following word must be kept static.

The most simple coding has a binary value one represented by some form of shift (up, down, left, right) and zero with no shift; more complex codings can obviously be generated. The small variation of the spaces between words or lines cannot be detected by the human eye but Brassil et al [12] show that the watermark can still be recovered after the tenth generation photocopy by electronically scanning the physical document and looking for the shifts.

The fine resolution of modern printers (commonly 300 and 600 dots per inch) can be used for modifying individual letters in the font used to print a document; this is known as character coding. For example, the letter ‘i’ can have the dot made larger, or its height increased; character modification equates to a binary one, a normal character equates to binary zero. In English text vowels will occur many times on each page and provide a large potential for coding schemes, this can easily provide a useful bandwidth on each page for carrying a watermark. Once again recovery of the watermark is achieved by scanning in a hard copy and analysing the characters.

Although these techniques are intended for hard copy output these processes could be used on electronic documents. It is common for archive material to be

electronically scanned and kept in a standard image format which can be manipulated.

4.5 Care needed in reviewing steganographic techniques

It is possible that a fraudster may wish to make an unauthorised copy of just part of the work-of-art. For example, a small part of a picture, or a line of dialogue from the sound track of a movie. For the watermarking process to be useful, the whole identity of the user needs to be embedded into every (or most) useful fragment of the work-of-art. It is possible that there will not be sufficient bandwidth available to do this. In some instances the owner of the work-of-art will only embed just a small part of the user’s identity in some fragments.

An important requirement of a watermark is that it cannot be removed accidentally or deliberately by a fraudster without significantly impairing the remaining work-of-art. If the watermark is embedded in a redundant part of the work-of-art then it can easily be removed by using compression algorithms or by conversion into another format. So ideally, watermarks should be embedded in the information of the work-of-art and not in the data that represents it. For example, publishers of four-figure tables have been known to put deliberate, but subtle, errors in some of their tables. Even if the tables are retyped by a fraudster wishing to resell the tables without authority, the original publisher can establish that it was their tables that have been copied.

To conclude we have shown that for each work-of-art there are different fragments that would need to be watermarked, there are different types of information to be protected and different levels of acceptability of fraud. Because of this each work-of-art needs to be analysed so as to decide which combination of watermark technology should be applied. It is important to note that there is no single watermarking scheme that is suitable for all works-of-art [15].

The discussion identified three requirements for a watermark or combination of watermarks applied to a work-of-art:

- a watermark should not intrude on the intended use of the work-of-art,
- a watermark should provide enough bandwidth to identify the authorised user of the work-of-art,
- removal of a watermark should degrade the work-of-art to a degree such that revenue lost by the owner (due to resale of the illegal copy) is minimal.

5. Proposed copy-protection scheme

5.1 Architecture

It is becoming increasingly common for Web pages to present information to users as a combination of text, audio and imagery. It would therefore be beneficial if copyright control could be given to each of the pages' individual elements. Web pages can be served to any type of computing architecture and so copy protection mechanisms must function on all computers with a minimum number of implementations. It is for these reasons that we feel the part of the copy protection system that resides on the users' machines would be best implemented as what we will call 'active components'.¹¹

The proposed scheme is as follows. The user accesses a page with some element that has copy protection. To recover the protected information the user's machine identifies itself to the webserver supplying the protected information. When the webserver is satisfied with the identity of the user's machine it supplies the information in a form that is bound to the user's machine so that only the active component on that machine can use it. The webserver will also recover any revenue that is due by interaction with an electronic payments scheme of the user's choice [16]. The active component is bound to the user's machine in a registration phase that is conducted when the user first downloads the component.

5.2 Registration phase

The active component is made unique to the user's machine during the registration phase (Fig 10).

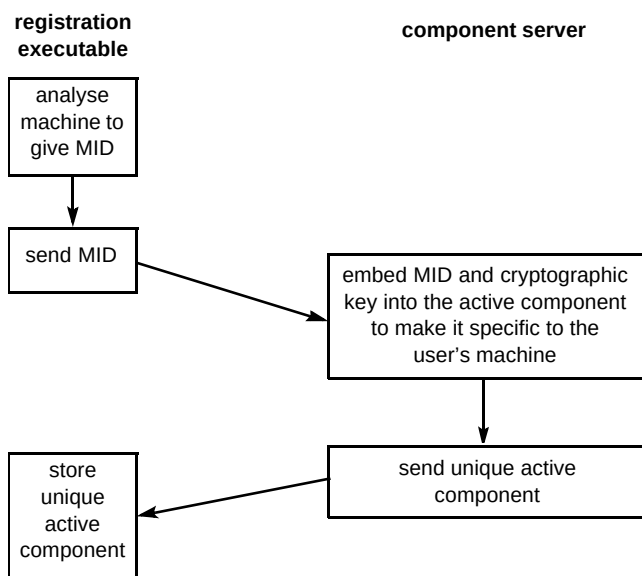


Fig 10 Registration protocol.

The user tries to access copy-protected information for the first time and discovers that they are required to have a specific active component, obtainable by downloading a registration executable from the component server (this is the machine that supplies the active components).

Once it has been downloaded the registration executable analyses the user's machine and generates a machine identifier (MID). The MID will be unique to the machine because it will be generated from all of the elements that make binding possible (see section 3.2).

The registration executable transmits this MID to the component server and awaits a response. The component server holds a partially complete version of the active component into which it embeds the MID and a unique cryptographic key to create an instance that is unique to the user's machine. The component server then sends the now complete active component to the registration executable¹².

The registration executable writes the active component to the hard drive of the machine to complete the registration phase and so it is now possible for the user's machine to try to access the protected material as explained in the next section.

5.3 On-line access to copy-protected material

The element that is protected may be a piece of text, an audio clip, or an image. To process each element the active component that is unique to the machine is activated for each. The process is detailed below and can be seen in Fig 11.

The active component analyses the user's machine and generates the MID, which it checks with the MID embedded within itself. When this check does not succeed the active component knows it is not running on the machine it was supplied to and does not try to recover the material from the webserver. When the active component is satisfied that it is running on the machine to which it was supplied, it sends the webserver a message stating the MID and the name of the file to be provided, and then awaits a reply.

The webserver replies with a message containing a randomly generated challenge to which the active component is required to give a response. To ensure the protocol is resistant to attack the response is computed as a cryptographic function of the MID, the randomly generated challenge, and the cryptographic key of the active component.

¹¹ An example of what we would call an active component is code implemented as a Java Applet, ActiveX Object or OpenDoc Part.

¹² It is important to note that the component server is not required to remember any part of the registration phase.

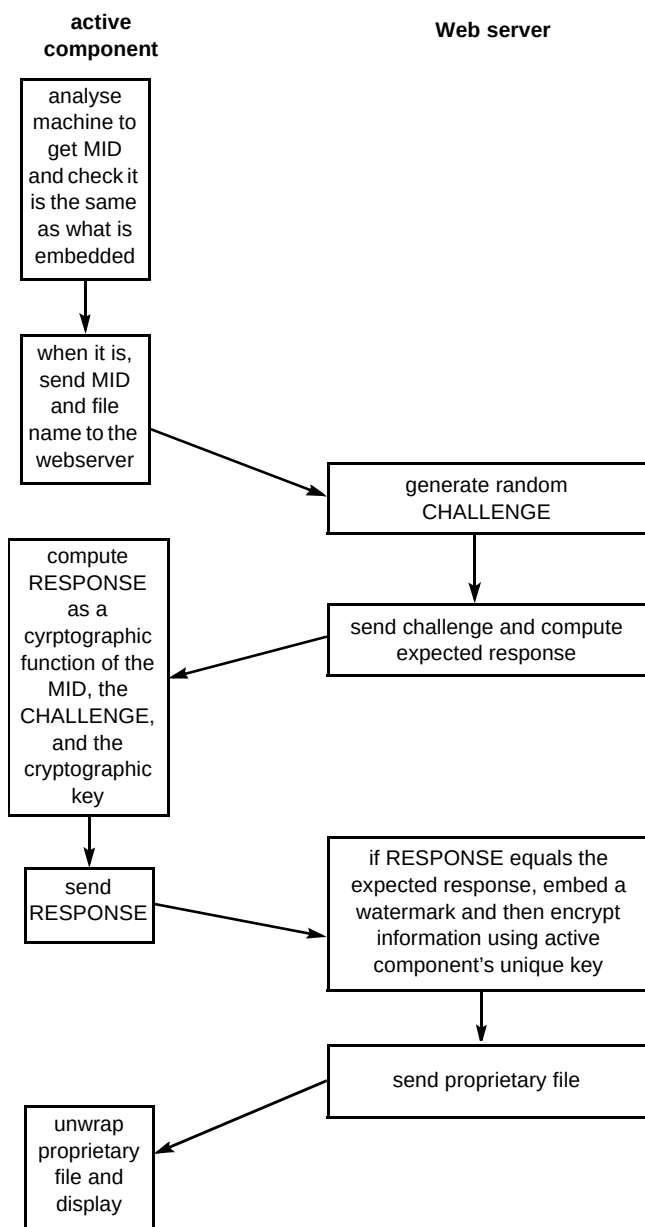


Fig 11 On-line access to copy-protected material.

Should the response from the active component be different to the one expected the webserver will refuse to serve the information. When the response is correct the webserver places a watermark into the information that indicates the user to whom the information was supplied and then encrypts it using the active component's cryptographic key. This is then sent to the active component using a proprietary file format.

Once the copy-protected information has been displayed by the active component the user may attempt to save it to long-term storage. The request to save will go to the active component which examines the webserver-supplied information to see if the action of taking a long-term copy is sanctioned. If so the active component saves the proprietary file to the hard disk for future off-line access.

5.4 Off-line access to copy-protected material

When the owner is prepared to supply an electronic copy to the user they wish to maintain control and so the information is saved in the proprietary file format. Only the active component executing on the user's machine can convert the information to a readable form, and so the active component must be capable of being used by many different types of application.

As an example we will take some text that has been supplied to a user by this copy protection mechanism and which he wishes to use as part of a report being written with a word processing application.

While using the word processing application he selects the proprietary file that contains the text he wants, and inserts it into his own document. This will invoke the active component and it performs the following actions. The active component analyses the user's machine and generates the MID, it checks this with the MID embedded within itself. When this check does not succeed the active component knows it is not running on the machine to which it was supplied and does not process the proprietary file. When the active component is satisfied that it is running on the machine to which it was supplied, it uses its cryptographic key to decrypt the proprietary file.

Once decrypted the content of the proprietary file can be put into the user's document if the controls held within permit such use. For example, an owner may permit users to hold long-term copies of their material but may not permit the material's reuse in other works-of-art.

6. Conclusions

This paper has shown that there are a number of requirements of a general-purpose copyright protection system that can be used in Internet-based services, and we have suggested mechanisms to satisfy these. There is a significant requirement for such a system to be integrated with emerging electronic payment schemes since these will provide the avenues for revenue collection [16]. It is very important that both the owners and users of any software-based system have assurance of the integrity of the software they are using. Software that is providing security operations has to be resistant to tampering and attack to ensure that works-of-art can only be used under the control of their owners. Because control cannot always be guaranteed there is a need for the application of robust watermarking techniques that meet the requirements we have stated.

References

- 1 Electronic Engineering Times, Issue 918 (September 1996).
- 2 Phillip V and Dodds W: 'The DVD invasion', DV Magazine, 4, No 11 (November 1996).
- 3 Tactile Marketing Group: 'DVD technology overview', http://www.tacmar.com/dvd_background.htm.
- 4 Phoenix S J D: 'Cryptography, trusted third parties and escrow' BT Technol J, 15, No 2, pp 45–62 (April 1997).
- 5 Denning D E: 'Cryptography and data security', Addison-Wesley (1982).
- 6 Choudhury A K, Maxemchuk N F, Paul S and Schulzrinne H G: 'Copy protection for electronic publishing over computer networks', IEEE Network Magazine (June 1994).
- 7 Adobe Acrobat overview, <http://www.adobe.com/prodindex/acrobat/main.html>
- 8 Meyer C D and Matyas S M: 'Cryptography — a new dimension in computer data security', John Wiley & Sons (1982).
- 9 Brin S, Davis J and Garcia-Molina H: 'Copy detection mechanisms for digital documents', Department of Computer Science, Stanford University (1994).
- 10 Schneier B: 'Applied cryptography', 2nd Edition, John Wiley & Sons (1996).
- 11 The Dice Company home page, <http://www.digital-watermark.com>
- 12 Brassil J T, Low S, Maxemchuk F and O'Gorman I: 'Hiding information in document images', AT&T Bell Laboratories (1995).
- 13 Low S H, Maxemchuk N F, Brassil J T and O'Gorman L: 'Document marking and identification using both line and word shifting', AT&T Bell Laboratories (1996).
- 14 Brassil J, Low S, Maxemchuk N and O'Gorman L: 'Electronic marking and identification techniques to discourage document copying', AT&T Bell Laboratories (1996).
- 15 Boland F M, Ruanaidh J J K O and Dautzenberg C: 'Watermarking digital images for copyright protection', in: 'Image processing and its applications', Conference Publication No 410, IEE (1995).
- 16 Putland P et al: 'Electronic payment systems', BT Technol J, 15, No 2, pp 32–38 (April 1997).

Bibliography

Quisquater J, Boucqueau J, Bruyndonckx O, Lacroix S and Macq B: 'Conditional access and copyright protection based on the use of trusted third parties', from the ACCOPI RACE project (1995).

Koch E, Rindfrey J and Zhao J: 'Copyright protection for multimedia data', Fraunhofer Institute for Computer Graphics, Germany (December 1994).



Ian Bramhill completed an HND in Computer Studies in 1983 and then worked for six years as a software engineer for military avionics companies.

In 1989 he joined BT working in the field of computer security. He was one of the first students to successfully graduate from the BT MSc course in Telecommunications engineering at London University.



Mathew Sims joined BT in 1991 after graduating from the University of Birmingham with a BSc (Hons) in Mathematics. While at BT he has designed and analysed cryptographic algorithms and protocols, designed cryptographic key management systems and analysed hardware and software for suitability for securing BT's systems.

He presently researches cryptography and information security to create new ideas for BT's products and services.